



RS003 Week 6 — English Worksheet

Topic: A Fleet of Ships — Many Parameters · **Course:** Pygame Space Shooter · **Time:** about 45 minutes

This week one function makes a whole **fleet of different ships**. You give `draw_ship` more parameters — position, colour, and size — so a single function can draw a small green scout, a big red boss ship, and a blue one, all at once.

Words to know: **parameter, argument, default value, keyword argument, scale (size)**

1 · Predict 🧠

Read each piece of code. Before you run it, write what you think will happen.

```
def draw_ship(x, body_color, size=1):  
    ...
```

This function takes three parameters. Which one has a default value, and what does that mean when you call the function?


```
draw_ship(200, GREEN, size=0.8)  
draw_ship(400, RED, size=1.2)  
draw_ship(600, BLUE, size=0.8)
```

Three ships are drawn. Which one is the biggest? Which one is in the middle?


```
w = int(40 * size)
```

If `size` is 1.2, what does `int(40 * size)` work out to? Why is `int(...)` needed?

2 · Run It 🏃

🎯 Type the example, run it, and compare the three ships

```
import pygame
pygame.init()

WIDTH, HEIGHT = 800, 600
screen = pygame.display.set_mode((WIDTH, HEIGHT))
clock = pygame.time.Clock()

SPACE = (10, 12, 40)
WHITE = (255, 255, 255)
GREEN = (60, 200, 120)
RED = (230, 60, 60)
BLUE = (80, 140, 255)

def draw_ship(x, body_color, size=1):
    w = int(40 * size)
    h = int(50 * size)
    nose = int(20 * size)
    y = HEIGHT - 120
    pygame.draw.rect(screen, body_color, (x - w // 2, y, w, h))
    pygame.draw.polygon(screen, WHITE, [(x, y - nose), (x - w // 2, y), (x + w //
2, y)])

running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False
    screen.fill(SPACE)
    draw_ship(200, GREEN, size=0.8)
    draw_ship(400, RED, size=1.2)
    draw_ship(600, BLUE, size=0.8)
    pygame.display.flip()
    clock.tick(60)

pygame.quit()
```

Describe the three ships — their colour, size, and position:

3 · Spot the Bug 🐛

Each block below was meant to do something but is broken. Fix it, then explain why the original was wrong.

Bug A — This should draw a ship in any colour, but every ship comes out the same colour no matter what you pass in.

```
def draw_ship(x, body_color, size=1):  
    pygame.draw.rect(screen, GREEN, (x - 20, HEIGHT - 120, 40, 50))
```

Hint: the function ignores its own `body_color` parameter.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — This should make a ship 1.2 times bigger, but Python crashes with an error about needing a whole number.

```
w = 40 * size  
pygame.draw.rect(screen, body_color, (x, HEIGHT - 120, w, 50))
```

Hint: `40 * 1.2` is a decimal. Pygame sizes must be whole numbers.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — You want most ships to stay normal size without typing `size=1` every time, but right now leaving size out causes an error.

```
def draw_ship(x, body_color, size):  
    ...  
  
draw_ship(400, RED)
```

Hint: give `size` a default value so it can be left out.

Write the fixed code (just the `def` line):

Why was it wrong? Why does your fix work?

4 · Modify It

Line up five ships in different colours

Call `draw_ship` five times across the screen, each with a different colour.

Write your five calls:

Make the middle one the boss

Use the `size` parameter to make one ship clearly bigger than the rest.

Write the call for your boss ship and the size you gave it:

Hint: a `size` above 1 makes it bigger; below 1 makes it smaller. Spread the X values evenly so they do not overlap.

5 · Make It 📷

Build your own ship fleet

Build a program that:

1. has a `draw_ship(x, color, size)` function,
2. draws several ships in a row,
3. uses different colours and at least two different sizes.

Send a **photo or video** of your fleet, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I added parameters for ...

I gave `size` a default value so that ...

My boss ship is bigger because ...

I used `int(...)` to ...

One tricky moment was when ...

If I had more time, I would ...

6 · Record Your Walkthrough 🎥

Take a video of your ship fleet. Talk through it like you are teaching someone. Try to use these words: **parameter, argument, default value, keyword argument, scale.**

1. Show your row of ships.
2. Read one function call out loud and name each argument.
3. Explain what the default value of `size` does.
4. Point at your boss ship and say what makes it bigger.

Write what you will say in your video. Plan it here before you record.

Submit ✓

Send this worksheet + your photo or video to teacher on KakaoTalk.